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Women, Demography, and Politics: How Lower Fertility Rates Lead to Democracy

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Abstract Where connections between demography and politics are examined in the literature, it is largely in the context of the effects of male aspects of demography on phenomena such as political violence. This project aims to place the study of demographic variables' influence on politics, particularly on democracy, squarely within the scope of political and social sciences, and to focus on the effects of woman-related demographics—namely, fertility rate. I test the hypothesis that demographic variables—female-related predictors, in particular—have an independent effect on political structure. Comparing countries over time, this study finds a growth in democracy when fertility rates decline. In the theoretical framework developed, it is family structure as well as the economic and political status of women that account for this change at the macro and micro levels. Findings based on data for more than 140 countries over three decades are robust when controlling not only for alternative effects but also for reverse causality and data limitations.

Keywords Fertility rate · Democracy · Demography and politics · Family structure

Introduction

As the meaning of this term in the Greek language suggests, *democracy* as a form of government stems from the δημος (Demos), the people. Accordingly, but largely outside of social science, scholars have studied the connections between demography and politics (Cincotta 2009; Engels 2010; Weber 2013). The first goal of this article is to place the study of the connections between demography and democracy squarely within political and social sciences, where it is "dramatically under-represented"

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(Goldstone et al. 2012:3). As demonstrated in this article, both theory and empirical evidence strongly suggest that demographic variables are not just affected by, but also affect, politics. Further, those effects are independent of variables traditionally studied as accounting for democracy, such as colonial legacies, economic growth, schooling, and globalization. Fertility is not simply influenced by economic conditions, such as labor force, but itself has a demonstrable impact on these conditions (in the case of Indonesia, for instance, see McDonald 2014), yet the consequence of fertility on politics and democracy has not been examined systematically.

Where the connections between demography and democracy were explored in the literature, the focus has been on male-related demographics. Thus, the second goal of this article is to shift the focus of analysis instead to demographic variables centered on women (e.g., Conzo et al. 2017; Flammang 1997; Tickner 2005; Tolleson-Rinehart and Carroll 2006). For instance, scholars have examined the effects of male youth bulges on political violence. Yet, demographic variables that are clearly related to women, such as fertility rate (the average number of children that a woman would have over her lifetime), have not received a comparable degree of scientific scrutiny.

The key argument is that the link between fertility rate and democracy is causal, with the former influencing the latter, and the effect being systematic, statistically significant, and substantively meaningful even when alternative accounts are controlled for. Four mechanisms mediate the effects of fertility rate on democracy: changes in family structure, economic conditions of women, political empowerment of women, and land policy. I test the hypothesis that demographic variables—female-related predictors, in particular—have an independent effect on political structure. In the theoretical path model, fertility influences the four intervening variables (changes in family structure, economic conditions of women, political empowerment of women, and land policy), which themselves link fertility causally to democracy and independent of control variables, such as economic conditions (GDP per capita) or schooling. The model tested is an abridged version of the theoretical model given that these intervening variables are not measured in the analyses.

Prior research has linked the political structure of society and demographic variables. Democracy, for instance, increases life expectancy, depresses infant mortality, and decreases fertility rates (e.g., Przeworski et al. 2000). However, there is also evidence in support of the opposite causal link from demography to politics. Population growth influences a range of political variables (Goldstone et al. 2012), such as political transitions (Organski et al. 1984), revolutions (Goldstone 1991), and participation in international conflicts (Cranmer and Siverson 2008). Demographic transitions that include changes in population age structure, fertility, mortality decline, urbanization, and life expectancy have also been linked to political shifts. As Dyson (2012:2) stressed, “these demographic processes comprise a major force behind the appearance and consolidation of democracy.” Yet, much of the research into the effects of demographics on politics has been published outside of political science. Furthermore, this scholarship has focused on right-hand variables (such as age structure and male youth bulges) and on left-hand variables (such as political violence). I argue that there is a distinct effect for fertility rate as a key right-hand variable on democracy, as the outcome variable.

Fertility decline is one of the most striking global phenomena of recent decades (Bongaarts 1999, 2002). During this period, many developing countries have undergone what is commonly referred to as their *first demographic transition*, in which fertility rates plummeted from six, seven, or even eight children per woman to replacement levels (of two per woman) or less, at which point they entered their *second demographic transition*. In 1950, the world's average fertility rate stood at 5.05, dropping to 2.49 in 2015. In the more-developed regions, the drop was modest from 2.81 to 1.68; however, in less-developed areas, the change was dramatic, dropping from 6.22 children per woman in 1950 to 2.62 six and one-half decades later.[1] In Asian countries, fertility rates dropped from 6.03 to 2.17 on average; in Latin America and the Caribbean, the drop was from 5.89 to 2.09.

Figure 1 outlines trends over time in levels of democracy and fertility rates in Indonesia from 1972 to 2008. Fertility rates (red diamonds) dropped from more than five children per woman to close to replacement levels during this period, a decline with momentous consequences for Indonesian society well documented in the literature (McDonald 2014).

From a political standpoint, this trend led to the fall of the New Order and the end of the Suharto regime after 31 years in power, ushering in the reform era in Indonesia (*Reformasi*). This era included a constitutional reform that introduced a system of separation of powers and checks and balances on the office of the president, which now had term limits and was required to deal with legislative institutions that were not the rubberstamp bodies they had been under Suharto. After the fall of Suharto, U.S. President Bill Clinton urged that Indonesia needed to begin "a real democratic transition" (Shenon 1998). The blue circles in Fig. 1, which indicate democratic conditions in the country on a scale of 0 (least democratic) to 10 (most) starkly reflect the political trend to which President Clinton referred.

Rather than deteriorating into a period of military dictatorship, Indonesia experienced substantial and consistent improvement in its democracy scores in the following years, a fact that has been attributed to declining fertility (Ananta 2006). Specifically, of the 14 ethnic groups in the country, those that played a key role in national politics and were instrumental in bringing about the fall of Suharto and the reform that followed were the Javanese (the largest group, constituting approximately 41 % of the population at the time) and the Madurese (the fifth largest, at slightly more than 3 % of the population). Although fertility rates among other groups were considerably higher (Sudanese, 2.4; Malays, 2.7; Bataks, 3.0), fertility rates for those two groups were below replacement level, at 1.8 for Javanese and 1.9 for Madurese—levels that had been consistent since the beginning of the 1990s. Indeed, it was these two groups that both showed the most positive attitudes toward democracy (Ananta et al. 2004) and were pivotal in steering their country in that direction. Given the projected decline in fertility rates for other ethnic groups as well, with the majority of the population in Indonesia projected by the United Nations to complete the first demographic transition by 2025, the prognosis for Indonesian democracy is positive.

Finally, the focus of our investigation here is comparative. Thus, it is not just the Indonesian case study that is of interest but rather how it compares with other cases. Indeed, the discrepancies in levels of democracy between Indonesia and countries in comparable positions in the 1960s can be at least partly ascribed to divergent trends in fertility rates. Declining fertility rate in Indonesia somewhat accounts for this country's

[1] Even excluding China, the drop in those countries (from 6.03 to 2.9) was comparable.

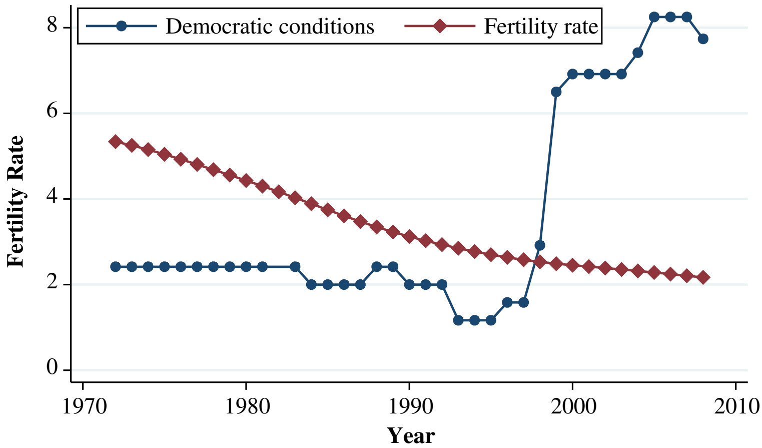

Fig. 1 Fertility rate and democracy in Indonesia. Data are from Quality of Government Database (https://qog.pol.gu.se/) and UN Population Division (http://www.un.org/en/development/desa/population/)

comparative edge in democratization (McNicoll 2011). Evidence for the link from fertility to democracy, as demonstrated shortly, is significantly stronger than what a particular case study or a specific pairwise comparison would suggest.

The key hypothesis wins unequivocal support in time-series cross-sectional analyses of data for more than 140 nations during a period of more than three decades. Apart from a highly significant and substantively meaningful statistical relation between fertility rate and democracy, this effect is also found when I control for traditional predictors of democracy. Furthermore, alternative demographic measures, albeit statistically significant in their own right, do not depress the main effect of fertility rate. This effect is robust to different measures of democracy, disparate databases, various model specifications, and varying measurement techniques. In conclusion, I discuss the implications for the study of democracy and for the relations between demographics and politics.

Theoretical Accounts linking Fertility Rate and Democracy

Theoretical accounts linking demographics and the political structure of society run the gamut from psychological to sociological and from evolutionary to economic theories. Some theories focus on psychological development: as a part of identity formation, young adults are thought to forgo their individuality in favor of group identity, possibly leading to political violence and the demise of a democratic order (Erikson 1968). Sociobiological approaches focus on competition over marital status and job security, and other theories discuss a societal bottleneck or the effects of a physiological neophilia, in which young individuals are willing to join their peer group in collective aggression (Choucri 1974). Likewise, behavioral ecological perspectives underscore the importance of male coalitional aggression in explaining such behavior, which increases the fitness of coalition participants (Mesquida and Wiener 1996). Indeed, institutional crowding may be one experience leading to violence among youth. Likewise, abundance of youth may lead to greater opportunity and lower costs in terms of violence (Urdal 2006).

In a marked departure from those theories, which are largely oriented toward the effects of male-related demographics on politics, here I focus on the political consequences of changes in fertility rate. As such, the theoretical framework developed here is focused instead on the effects of women-related demographic variables on politics (Davis and van den Oever 1982; Munck and Verkuilen 2002). This contribution is important because it examines the question within a framework that goes beyond and expands on the male-focused research in this field thus far (Tickner 2005).

Four mechanisms by which low levels of fertility rate contribute to the establishment of democracy are proposed here: changes in family structure, economic conditions of women, political empowerment of women, and land policy. Those mechanisms pertain to both the macro and micro levels.

Lower fertility entails changes in family structure, which lead to changes in political behavior in a way that is conducive to a democratic form of government (Breyer and von der Schulenburg 1990). The rising cost and diminishing economic value of children in a society with declining fertility rates lead to smaller families (Bongaarts and Watkins 1996). Put simply, family structure changes when it is no longer necessary to have many children as an insurance policy against infant mortality and when there is no need to guarantee a sufficient number of working hands for the family by increasing its size. Families can use the resources freed up by lower fertility toward educating their children (Galor and Zang 1997).

Furthermore, in polities with smaller families, not only are children better educated, but levels of uncertainty of future events decrease. Thus, individuals are well informed, live in a more predictable environment, and are more likely to plan their future and are suited to do so. Hence, they are more likely overall to organize and articulate their interests, fight for their needs, and claim their rights. Changes in family structure therefore go beyond the size of the family; indeed, via their influence on family structure, lower fertility rates lead to more political participation, articulation of interests, and claiming of rights (Lee 1976).

In sum, with declining fertility rates, the structure of the family changes, which results in improved education, predictability, and planning for the future. These, in turn, will boost forms of political behavior that, by definition, are parts of a more general process of democratization.

Lower fertility rates lead to democracy through their crucial effect on women's role in the economy and in political life. On the macro economic level, fertility has been shown to influence several parameters (Bleaney and Nishiyama 2002). High fertility among certain groups, such as the poor, leads to lower education and thus to limited economic growth (de la Croix and Doepke 2003).

More specifically to women, research indicates a connection between women's reproductive choices and structural obstacles in employment (Huckle 1981). With lower fertility, women are better educated and are more likely to hold salaried jobs and take an active part in economic activity (Huckle 1981).

As for political empowerment for women, fertility rates influence both elites and the masses. For elites, childrearing can act as an obstacle to becoming political decision-makers when household responsibilities are gendered and women find it difficult to imagine balancing household duties and paid work (Lawless and Fox 2010; Lee 1976). Fertility rates, thus, have a direct influence on elites. Countries where fertility rates are high have fewer women in leadership positions. Lawless and Fox (2010:2) found that "gender exerts one of the strongest influences on who ultimately launches a political career," arguing that socialization concerning gender roles and childcare hinder women from seeing themselves as political candidates.

One key way in which family affairs may be independent from economic variables in influencing politics (Iversen and Rosenbluth 2010) is the decision to be politically active. If socialization and social roles restrict women's perceptions of their ability to be politically active and to take an active role in this arena, economic changes will be of lesser consequence. Gender itself has been demonstrated to influence both interest in and willingness to be politically active (Lawless and Fox 2010; Sapiro 1981, 1984). The strongest evidence for this comes from affluent societies, where women are significantly less likely to enter politics, despite their greatly increased household bargaining power compared with women in less-prosperous societies (Iversen and Rosenbluth 2010). In richer nations, women's own economic conditions and the general economic reality may putatively be conducive to more engagement with politics either in terms of political participation or actually in terms of running for office. Dramatic changes in fertility rates have an effect that works at both the macro and micro levels. Even if social expectations and gender roles change minimally, a reduction in childbearing will leave women with considerably more available resources to consider economic activity, social engagement, and political participation in various forms. Thus, it is fertility rates that will influence individual decisions about political engagement and at the aggregate level will influence social and political realities.

Some of the processes that start with reduced fertility may cyclically boost the effect toward better representation, better political inclusion, and thus improved democratic conditions. For instance, if reduced fertility allows even a limited number of women to run for office, those women would be more likely to address women's issues, including representing women's policy preferences and enhancing political participation among female citizens as a group (e.g., Gerrity et al. 2007; Swers 2013). These politically engaged women serve as role models for those who otherwise may choose not to be involved in politics (Wolbrecht and Campbell 2007, 2017) and introduce a different style of leadership that may be more appealing to other female citizens (Tolleson-Rinehart 1991).

The link between reduced fertility and democracy exists at the micro level of the individual as well. Echoing Maslow (1943), a mother of seven in 1950 was likely preoccupied with fending for the physiological needs of her offspring, leaving little time to consider the social, economic, and political reality of the nation she lived in. Conversely, to allow her two children in 2017 to fulfill their self actualization needs (Maslow 1943), a mother in the early twenty-first century is more likely to have the available resources to consider the type of society and political reality for which she aspires. In the early twenty-first century, children are less a form of insurance for the sustenance of the family and more a means for self-actualization (their own as well as their mothers'). These factors at the individual level increase the likelihood of active incorporation into the political sphere. When women are more strongly motivated to run for office, it would be reflected in levels of political participation at the level of both the masses and the elites.

At the individual level, gender can decrease the likelihood of political participation, with women participating less than men. Yet, education and access to information play a critical role in increasing political participation, particularly for disadvantaged groups, information-poor societies, and the poor. Lower fertility rates free up resources needed for women to become politically relevant: women become better educated, gain experience in the salaried labor force, and train in professions that dominate politics, such as law (Inglehart and Norris 2003). The effect of education and information in increasing levels of political participation is so robust that it is found even in the absence of economic growth or reduction of poverty (Inglehart and Norris 2003). Lower fertility rate is also often associated with an increase in the average age of women at first birth, a factor with positive implications for educational attainment, employment opportunities, and political incorporation (Brunnbauer 2000). A woman who has her first child at an older age is more likely to be politically active later in her life. In sum, having fewer children, on average, women gain the requisite experience and education, becoming empowered politically and economically. Consequently, political inclusion and interest articulation rise, leading the polity as a whole to democratize.

Although fertility rate has been linked to politics in other research as well, the causal pathways suggested in the theoretical framework developed here are novel. In previous theories, the links from fertility to democracy (or to any other political variable for that matter) are mediated by other demographic variables (Davis and van den Oever 1982; Garrard et al. 2000; Lee 2003; Munck and Verkuilen 2002). Conversely, the key argument here is that there is a link from fertility to democracy. Further, the empirical tests used here are markedly different from those offered for extant theories. Fertility rate is usually not specified but is subsumed indirectly—for instance, by its effects on population age structure or median age (Dyson 2012). Here, fertility rate is the key predictor.

Last, from a political economic perspective, changes in land ownership (establishment of clearly defined property rights) may also be crucial. This last theoretical account, however, is offered here as a tentative explanation given that the scope of this article does not allow a full consideration of its complexities. When the average number of children per family drops over the course of three decades from approximately seven to less than three, changes in land allocation will also take place, amounting to land reform in some cases (Joireman 2007).

The consequent legal formalization of clearly defined and enforced property rights allows the transformation of assets into capital, a process that contributes to both economic and political development (de Soto 2000). With clearly defined property rights, the future is more predictable to individuals, groups, and organizations. The ability to turn assets into capital side by side with this new predictability will lead to increased economic activity, with contractual obligations being more likely, and to political development through the creation of political groups, unions, and organizations with shared interests.

Economic and political development due to changes in land policy and in the formalization of property rights leads the country invariably in the direction of democracy (Fukuyama 2004). As fertility rates drop, reforms in land policy and consequent formalization of property rights increase the likelihood of democratization.

Thus, the first key hypothesis is as follows:

> *Hypothesis 1:* As fertility rate decreases, the country democratizes, even when traditional predictors of democracy are controlled for.

In addition, although the links between male-related demographic predictors (such as median age and youth share) and political variables (such as political violence and democracy) are well established (Cincotta 2009; Urdal 2006; Weber 2013), the aforementioned causal pathways indicate a link from fertility rate to democracy. Therefore, the effect of fertility rate should be significant above and beyond the effects of other demographic variables. The second key hypothesis is therefore as follows:

> *Hypothesis 2:* Fertility rate influences democracy beyond the effects of other demographic variables, such as median age and youth share.

Antecedents of democracy that have been the focus of literature in political science go well beyond demographic variables and range from legitimacy to economic development (Lipset 1959) to schooling (Barro 1999). To control for alternative accounts, in addition to fertility rate, the following effects are also tested: GDP per capita, level of globalization, French and British colonial heritages, the size of the Muslim constituency, schooling (Barro 1999), other democracies in the region, percentage of women using contraceptives, consumer price index (CPI), and ethnolinguistic fractionalization. Economic growth has been linked to competitive politics and democracy (Dahl 1971), and economic modernization influences democratic stability and the process of democratization itself (Bollen and Jackman 1985; Feng and Zak 1999; Hadenius 1994). As GDP per capita increases, a concurrent rise in democracy is expected (Przeworski et al. 2000). *Ceteris paribus*, the expectation is that as a country globalizes, it will also become more democratic. Highlighting the role of British colonialists in building democratic institutions in their colonies, Lipset (1959) and others contended that British colonial heritage should increase the likelihood of a democratic government. Accordingly, although French colonial heritage may decrease the likelihood of a democratic form of government, British colonial heritage is expected to have the opposite effect (Przeworski et al. 2000). Last, religion may influence reproduction (Forman-Rabinovici and Sommer 2018). In particular, the size of the Muslim constituency is relevant not only because it influences the status of women (Callaway 1987) but also because the fertility model is somewhat different in Muslim countries (Obermeyer 1992). Furthermore, the weak distinction between the religious and political communities in Muslim countries may also stand in the way of building firm democratic institutions (Huntington 1991). Therefore, with a larger Muslim constituency, the likelihood of a democratic form of government decreases.

Measures of Democracy

Most of the existing indicators of democracy measure Dahl's two underlying dimensions of polyarchy: contestation and inclusiveness (Coppedge et al. 2008; Munck and Verkuilen 2002). The elements of contestation typically include elected offices, the nature of the elections, electoral contestation, free and fair elections, freedom of organization, and expression and pluralism in the media (Alvarez et al. 1996; Hadenius 1992). As for inclusiveness, to reflect the "proportion of the population entitled to participate on a more or less equal plane in controlling and contesting the conduct of government" (Dahl 1971:2), the measures used are usually adult suffrage, the political rights of women, and levels of political participation (Coppedge et al. 2008). Furthermore, to measure contestation, objective indicators of electoral and institutional competition are used (Gervasoni 2010). Finally, the protection of rights and liberties also indicates the level of democracy (O'Donnell 2004). To measure democratic conditions, the key outcome variable specified in this study is the polity score (ranges from –10 to 10), which is largely based on the aforementioned theoretical and measurement approaches to gauge democracy. To test for robustness, I specify freedom house score and an index combining the polity score and the freedom house score (ranges from 0 to 10) as dependent variables in some of the models. For an additional test of robustness, some models are also estimated with the Vanhanen democratization index as a dependent variable. The results for the latter models are reported in a separate table.

Reverse Causality

Apart from its anticipated influence on age composition and population growth (Coale 1986), fertility has been shown to influence society and politics more broadly (Grant et al. 2006). Indeed, changes in fertility rates were pivotal in the advancement of certain political agendas, most notably that of socialism (Winter 1988). Furthermore, fertility rates influence the structure of the national system of education (de la Croix and Doepke 2009), parameters of social stability (Chesnais 1998), and decision-making authority in both the family and the community (Dasgupta 1995). In this sense, the hypothesized causal link between fertility and democracy is in line with existing scholarship. Yet, the opposite causal direction from a democratic form of government to various demographic variables has been extensively studied as well. Democracy, for instance, reduces the infant mortality rate (Ross 2006). More specific to fertility rate, democracies (because of their pluralism) are less likely to adopt rigorous pronatalist policies (McIntosh 1981); indeed, the more democratic the country, the lower its fertility rate (Beer 2009). Likewise, levels of democracy influence women's well-being, with fertility being one of its indicators (Wejnert 2003). As Przeworski et al. (2000) suggested, regime type influences the growth of the population even more than it influences economic variables. Thus, the direction of causality between the predictor (fertility rate) and the outcome variable (democracy) is an issue. To test reverse causality, I estimate models with lagged variables, discussed further in the Methods and in the Results sections.[2]

Data and Methods

Democratic conditions are measured based on the qualities of executive recruitment; constraints on executive authority; political competition; freedoms of expression and belief; associational and organizational rights; rule of law; and free participation in the

[2] Within the context of democracy, modernization, and fertility, finding good instrumental variables for the estimation of two-stage least squares (2SLS) models is not feasible.

political process, including the right to vote in legitimate elections, compete for public office, join political parties and organizations, and elect effective and accountable representatives. The dependent variable is the Polity scale of –10 to 10 in Tables 1, 3, and 4, and Table 6 in the appendix. To test for robustness, Table 2 includes the Vanhanen index as the outcome variable. The outcome variable used in Table 5 is the Freedom House scale ranging from 0 to 10 and averages Freedom House and Polity. This average index is an imputed variable that tests robustness well because it performs better both in terms of validity and reliability than its constituent parts (Hadenius and Teorell 2005).

To measure fertility rate, I use data from the gapminder project[3] and from the U.N. Population Division World Population Prospect.[4] On top of fertility rate, two additional demographic variables are included: median age in the population, and youth share. As median age in the population increases, we also expect democratic conditions to improve.

Different scholars have measured youth share in a variety of ways. In some cases, the age range may be 20–45 (e.g., Moller 1968: Table 1); others use a younger cohort, such as 15–29 or 20–39 (e.g., Weber 2013). Furthermore, Urdal (2006) postulated a demographic dividend, where because of a dependency burden, the effect of a youth bulge may be contingent on the size of the cohort that it precedes. The latter is a direct function of fertility rate. To specify this lowered dependency burden due to levels of fertility, which may influence the demographic dividend, youth share is measured as the share of 20- to 39-year-olds in the population. It is specified as a main effect and is also interacted with fertility rate. Data for both median age and youth share are taken from the gapminder project[5] (see also Cincotta and Doces 2012; Urdal 2012).

Based on the CIA fact book,[6] *percentage Muslim* indicates Muslims as percentage of the population. Another measure for Muslims in the population comes from a report by the Pew Forum on Religion and Public Life (2009), which provided up-to-date and comprehensive demographic estimates of the number of Muslims in countries and territories for which the U.N. Population Division provided general population estimates.[7]

To measure international connections, I use the KOF Indexes of Globalization (Dreher 2006; Dreher et al. 2008). The indexes for the globalization variables are measured in line with the standard in the discipline and range from 0 to 100. This is a weighted average of social, political, and economic globalization. GDP per capita is reported in constant U.S. dollars at base year 2000 (Gleditsch 2002).

[3] Information on this project can be found online (gapminder.com).

[4] These data are also available online (https://esa.un.org/unpd/wpp/Download/Standard/Interpolated/).

[5] For further discussion of measurement issues, see Urdal (2011).

[6] The CIA fact book is available online (https://www.cia.gov/library/publications/the-world-factbook/).

[7] This study counts all groups and individuals who self-identify as Muslim, which allows comparative analyses between countries. The number of Muslims is calculated by multiplying the United Nations 2009 total population estimate for each country by the single most recent and reliable estimate of the percentage of Muslims there. This calculation is based on the assumption that Muslim populations are growing at the same rate as each country's general population, which is a conservative assumption. Sources include national censuses, demographic and health surveys, and general population surveys and studies.

Table 1 Generalized estimating equation model predicting the effect of fertility rate on democratic conditions with polity score as outcome variable: A time-series cross-sectional analysis, 1972–2006

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Fertility Rate	–1.5***	—	–1.44***	—	–1.6***	–2.54***	–2.33***	–2.24***
	(0.199)		(0.21)		(0.2)	(0.44)	(0.44)	(0.44)
Median Age	—	0.166***	0.05	—	—	—	0.09*	0.053†
		(0.039)	(0.037)				(0.04)	(0.039)
% Youth	—	—	—	0.032	–0.056*	–0.18**	–0.19**	–0.17**
				(0.036)	(0.030)	(0.07)	(0.07)	(0.06)
Fertility × % Youth	—	—	—	—	—	0.033*	0.03*	0.03*
						(0.014)	(0.014)	(0.014)
GDP (per capita)	–0.000009	–0.000005	–0.000015	0.0000061	–0.000017	–0.000021	–0.00002	–0.000043*
	(0.000015)	(0.000016)	(0.000014)	(0.000019)	(0.000015)	(0.000015)	(0.00004)	(0.000019)
Globalization	0.046**	0.072***	0.041*	0.089***	0.046*	0.047*	0.043*	0.041*
	(0.017)	(0.016)	(0.018)	(0.017)	(0.019)	(0.019)	(0.019)	(0.019)
History as a French Colony	1.49†	0.225	2.07*	–0.99	1.9*	1.93*	2.36*	1.42†
	(0.88)	(0.96)	(0.95)	(1.01)	(0.9)	(0.96)	(0.99)	(1.01)
Legal Origin in British Common Law	3.15**	1.92†	3.64***	0.616	3.48**	3.48***	3.9***	3.43***
	(1.03)	(1.13)	(1.12)	(1.15)	(1.1)	(1.09)	(1.1)	(1.07)
Percentage of Muslims in the Population	–0.056***	–0.071***	–0.06***	–0.078***	–0.06***	–0.057***	–0.05***	–0.05***
	(0.015)	(0.012)	(0.012)	(0.012)	(0.01)	(0.01)	(0.01)	(0.01)
Other Democracies in the Region	—	—	—	—	—	—	—	4.23***
								(1.29)
Constant	5.3***	–4.8**	3.6*	–1.5	6.9***	10.55***	8.16***	7.19**
	(1.53)	(1.58)	(2.08)	(1.6)	(1.7)	(2.29)	(2.5)	(2.5)

Table 1 (continued)

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
N	4,310	4,025	4,025	3,772	3,772	3,772	3,737	3,737
Number of Groups	139	127	127	127	127	127	126	126
Observations per Group								
Minimum	3	3	3	7	7	7	7	7
Average	31	31.7	31	29.7	29.7	29.7	29.7	29.7
Maximum	35	35	35	35	35	35	35	35
Wald χ^2	6	6	7	6	6	8	9	10
df	292.8	198.86	309.17	119.2	329.5	318.43	383.5	488.94
Prob > χ^2	.000	.000	.000	.000	.000	.000	.000	.000

Note: Robust standard errors, clustered on the nation, are shown in parentheses.

$^{\dagger}p < .10$; $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$ (one-tailed tests, where directionality is hypothesized)

Table 2 Generalized estimating equation model predicting the effect of fertility rate on democratization with Vanhanen democratization index as outcome variable: A time-series cross-sectional analysis, 1972–2006

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
Fertility Rate	–2.16***	—	–1.9***	—	–2.5***	–3.23**	–2.41**
	(0.41)		(0.4)		(0.37)	(0.96)	(0.84)
Median Age	—	0.33***	0.17*	—	—	—	0.25**
		(0.09)	(0.08)				(0.09)
% Youth	—	—	—	–0.12†	–0.27***	–0.36*	–0.39*
				(0.07)	(0.07)	(0.15)	(0.15)
Fertility × % Youth	—	—	—	—	—	0.025	0.023
						(0.032)	(0.029)
GDP (per capita)	0.0001*	0.00009*	0.00009*	0.00013*	0.00012*	0.00011*	0.000019
	(0.00005)	(0.00005)	(0.00005)	(0.00006)	(0.00005)	(0.00005)	(0.000038)
Globalization	0.13***	0.167***	0.12***	0.18***	0.127***	0.127***	0.106***
	(0.029)	(0.026)	(0.029)	(0.026)	(0.028)	(0.028)	(0.027)
History as a French Colony	–0.25	–1.56	0.57	–4.1*	0.175	0.176	–1.4
	(1.9)	(1.97)	(2.01)	(1.89)	(1.87)	(1.86)	(1.6)
Legal Origin in British Common Law	–0.25	–1.37	0.66	–4.04*	0.322	0.31	0.4
	(2.08)	(2.11)	(2.16)	(1.98)	(2.05)	(2.01)	(1.82)
Percentage of Muslims in the Population	–0.09***	–0.11***	–0.089***	– 0.11***	–0.08***	–0.08***	–0.03*
	(0.015)	(.014)	(0.014)	(0.015)	(0.014)	(0.014)	(0.01)
Other Democracies in Region	—	—	—	—	—	—	14.02***
							(2.12)
Constant	17.14***	0.26	11.66**	12.8***	26.05***	28.7***	17.08***
	(2.83)	(3.3)	(3.97)	(3.06)	(3.12)	(4.85)	(4.35)
N	4,683	4,310	4,310	4,019	4,019	4,019	3,949
Number of Groups	151	136	136	138	138	138	136
Observations per Group							
Minimum	12	13	13	7	7	7	7
Average	31	31.7	31.7	29.1	29.1	29.1	29
Maximum	35	35	35	35	35	35	35
Wald χ^2	6	6	7	6	7	8	10
df	293.91	260.47	274.62	220.62	377.01	378.05	740.65
Prob > χ^2	.000	.000	.000	.000	.000	.000	.000

Note: Robust standard errors, clustered on the nation, are shown in parentheses.

$^{\dagger}p < .10$; $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$ (one-tailed tests, where directionality is hypothesized)

Data are cross-sectional time series, listing all countries for a period of approximately 30 years. I estimate generalized estimating equation (GEE) models (see Zorn 2001). A marginal approach, such as the GEE, is appropriate in this case because the focus here is on the variables influencing democratic conditions in the country rather than the effects in a particular country, for which a conditional approach would suffice (Zorn 2001). Due to the aforementioned data structure, I employ a GEE model with a first-order autoregressive component. All models are estimated with robust standard errors. For a list of countries included in the multivariate analyses, please refer to the appendix.

Results

The results in Table 1 lend strong support to the key hypotheses. As Model 1 indicates, a strong and highly significant negative connection exists between fertility rate and democracy; *ceteris paribus*, for each additional child per woman on average, the democracy measure drops by 1.5 units. This effect is statistically significant at the highest levels of significance, controlling for alternative hypotheses. Taking the example of Mexico, the fertility rate in this country between 1972 and 1998 plummeted from 6.73 to 2.71—a 4.02 decrease over the 26-year period. During this same period, democracy rose from a level of –6 on the Polity scale to a level of 6, an overall increase of 12 units. The 4.02-unit drop in fertility rate contributed approximately 6 of the 12-unit increase in the democracy index. In other words, 50 % of the improvement in democratic conditions in Mexico in this period is accounted for by the decrease in fertility rate. Much of the democratic shift leading up to the historical elections in 2000 of the first non-PRI (Partido Revolucionario Institucional) president since 1929 in Mexico is attributable to changes in fertility rates in this country in the preceding decades.

Model 1, thus, lends unequivocal support for Hypothesis 1 (the effect of fertility rate on levels of democracy is independent of the effects of traditional predictors of this form of government). Yet, as prior research has indicated, other demographic variables influence political reality. Further, because demographic variables are related to each other in various ways, I include additional models to test Hypothesis 2, which states that the effect of fertility rate on levels of democracy should stand even when other demographic variables—particularly those concerning population age structure—are controlled for. Models 2–7 in Table 1 examine the effects of median age, youth share, and the interaction of youth share with fertility. Because of a dependency burden, I specify an interaction between the share of youth in the population and fertility rate—what Urdal (2006:611) identifies as a *demographic dividend*.

Models 6–8 indicate that *ceteris paribus*, fertility rates influence levels of democracy in a highly significant manner, even when other demographic variables and possible interactions between fertility and those demographic variables are controlled for. Share of youth in the population also has an effect: as the share of youth increases, levels of democracy diminish. I find an interaction between fertility rate and the share of youth in their effects on democratic conditions. The effects of fertility rates are less pronounced for the first generation in which such a pattern appears. In other words, after long periods of high fertility, which created a large youth share in the population at the present time, the effect of fertility on politics is somewhat diluted. Yet, this interaction term indicates a contingent effect that is only an increment. The main effect of fertility, which is still highly significant and approximately of the same substantive magnitude across Models 6–8, indicates that on top of this contingent increment and on top of the main effect for youth share, fertility rate has an independent and substantively meaningful effect on levels of democracy. The results are substantively indistinguishable in Model 8 in Table 1, where all demographic variables and the share of democracies in the region are specified.

Table 2 provides a similar set of multivariate models. Models estimated here, however, specify Vanhanen's democratization index as the outcome variable. The results remain largely unchanged, which indicates robustness. Model 1 lends unequivocal support to Hypothesis 1. The democratization index ranges from 0 to 49, which accounts for the different size of the coefficient on fertility rate, compared with Table 1. One additional child per woman decreases the level of democratization by approximately 2 units, which is comparable to a 1.5-unit drop on the Polity scale that ranges from –10 to 10. In addition, as Models 2–6 indicate, this effect is statistically significant and substantively meaningful even when median age and youth bulge are controlled for. Other than the interaction term, results for Models 6 and 7 in Table 2 largely resemble the findings for the respective models in Table 1, with fertility being highly significant and in the anticipated direction.

The results for the control variables are as anticipated. As the country globalizes, its form of government is likely to become more democratic. Whereas being a former French colony is inconsequential in several of the models, having legal origins in British common law increases the level of democracy. With a greater share for a Muslim constituency in the nation, the country is less likely to be democratic. This effect is highly significant and makes the earlier argument about the effects of fertility on democracy in the case of Indonesia (the largest Muslim country in the world) even more noteworthy.

Controlling for Reverse Causality

To examine reverse causality, I replace the fertility variables in Tables 1 and 2 with lagged values for fertility. This model specification is one solution for the possible causality issue between democracy and fertility, testing the effects of fertility rates several years back on democratic conditions in the country at present.[8] Statistically significant relations between lagged fertility variables to democracy would seem to indicate that changes in fertility (3, 5, 18, or 36 years ago) influenced democracy. This logic is strongly supported, particularly when alternative effects of other predictors are controlled for in the same models.

Table 3 presents five models specifying fertility variable lagged 1, 3, 5, 18, and 36 years. Models 4 and 5 test the current effect on democratic conditions of fertility rate at the time the voters, who are coming of age at present, were born (Model 4) or two political generations ago (Model 5). Support for the theory in all models in Table 3 is unmistakable. The effect of fertility is significant at the .0001 level. To examine the robustness of the tests for reverse causality, I also estimated the five models in Table 3 with the Vanhanen democratization index as the dependent variable. The results (not reported here) are substantively indistinguishable.

The data for the models in Table 4 come from an independent source (Alvarez et al. 1996). If the results remain unchanged in analyses based on data from an alternative source, this would be another indicator of robustness. This model also controls for other variables that the literature suggests may influence processes of democratization: ethnolinguistic fractionalization and the presence of other democracies in the region

[8] Although other approaches such as a two-stage least square (2SLS) model are also potential solutions, a 2SLS model is unfeasible in the substantive context of this article. The power of such a model would be considerably limited because of the inability to find good instrumental variables in a model that already specifies independent variables related to development, age structure, and fertility, and where the outcome variable is democracy or democratization.

Table 3 Controlling for reverse causality using lagged values for fertility

Variable	Model 1 (fertility lagged $t-1$)	Model 2 (fertility lagged $t-3$)	Model 3 (fertility lagged $t-5$)	Model 4 (fertility lagged $t-18$, one "political generation")	Model 5 (fertility lagged $t-36$, two "political generations")
Fertility Rate Lagged ($t-1$)	–1.02***	n/a	n/a	n/a	n/a
	(0.20)				
Fertility Rate Lagged ($t-3$)	n/a	–1.17***	n/a	n/a	n/a
		(0.21)			
Fertility Rate Lagged ($t-5$)	n/a	n/a	–1.43***	n/a	n/a
			(0.24)		
Fertility Rate Lagged ($t-18$)	n/a	n/a	n/a	–1.08***	n/a
				(0.22)	
Fertility Rate Lagged ($t-36$)	n/a	n/a	n/a	n/a	–0.74***
					(0.21)
GDP (per capita)	0.00001	–0.000003	0.00001	–0.00005	–0.000014
	(0.00002)	(0.000024)	(0.00002)	(0.00003)	(0.000057)
Globalization	0.076***	0.06***	0.076***	0.13***	0.08**
	(0.017)	(0.02)	(0.021)	(0.021)	(0.02)
History as French Colony	0.7	1.6	1.7*	1.6*	2.6*
	(0.99)	(0.98)	(1.9)	(0.8)	(1.05)
Origin in British Common Law	2.07*	2.55*	2.9**	2.26*	1.06
	(1.07)	(1.11)	(1.09)	(0.94)	(1.1)
% Muslims	–0.06***	–0.06***	–0.054***	–0.059***	–0.091***
	(0.012)	(0.012)	(0.012)	(0.013)	(0.014)
Constant	2.83	3.7*	3.76*	1.17	3.69*
	(1.58)	(1.85)	(1.99)	(2.08)	(2.01)

Table 3 (continued)

Variable	Model 1 (fertility lagged $t - 1$)	Model 2 (fertility lagged $t - 3$)	Model 3 (fertility lagged $t - 5$)	Model 4 (fertility lagged $t - 18$, one "political generation")	Model 5 (fertility lagged $t - 36$, two "political generations")
N	2,502	2,444	2,354	1,806	635
Number of Groups	133	133	133	135	129
Observations per Group					
Minimum	7	5	5	4	2
Average	18.8	18.4	17.7	13.4	4.9
Maximum	34	35	35	28	10
Wald χ^2	6	6	6	6	6
df	218.4	223.94	269.97	376.26	167.96
Prob > χ^2	.000	.000	.000	.000	.000

Note: Robust standard errors are shown in parentheses.

*p < .05; **p < .01; ***p < .001

Table 4 Robustness test

Variable	Alvarez et al. Data Set + Additional Controls, Including Cumulative Years of Education of the Average Member of the Labor Force	Alvarez et al. Data Set + Additional Controls, Including Central Government Expenditure on Education as Share of GDP
Fertility Rate	–0.31**	–0.32**
	(0.13)	(0.13)
GDP	0.000027	0.00002
	(0.000032)	(0.00003)
Social Globalization	0.02*	0.026*
	(0.01)	(0.013)
Economic Globalization	–0.003	–0.002
	(0.01)	(0.011)
Political Globalization	–0.0004	–0.0013
	(0.009)	(0.0091)
History as French Colony	–0.53	–0.55
	(0.49)	(0.49)
Origin in British Common Law	1.18**	1.17**
	(0.4)	(0.4)
% Muslims	–0.002	–0.0021
	(0.004)	(0.0045)
Ethnolinguistic Fractionalism	0.12*	0.12*
	(0.06)	(0.06)
Other Democracies in the Region	4.4***	4.4***
	(0.88)	(0.9)
% Women Using Contraceptives	0.00010	0.0011†
	(0.0007)	(0.0007)
Consumer Price Index	0.0002***	0.00022***
	(0.000058)	(0.000058)
Cumulative Years of Education of the Average Member of the Labor Force	–0.009	—
	(0.008)	
Central Government Expenditure on Education as Share of GDP	—	0.0006
		(0.012)
Constant	3.7**	3.7**
	(1.2)	(1.2)
N	1,057	1,057
Number of Groups	105	105
Observations per Group		
Minimum	3	3
Average	10.1	10.1
Maximum	18	18
Wald χ^2	13	13
df	872.16	874.07
Prob > χ^2	.000	.000

Note: Robust standard errors are shown in parentheses.

$^{\dagger}p < .10$; $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$

Table 5 Robustness tests controlling for average years of schooling

Variable	Model 1 Females Over 15	Model 2 Females Over 25	Model 3 Males Over 15	Model 4 Males Over 25	Model 5 Males and Females Over 15	Model 6 Males and Females Over 25
Fertility Rate	–3.2**	–3.04**	–3.37**	–3.13**	–3.3**	–3.04**
	(1.17)	(1.17)	(1.17)	(1.17)	(1.17)	(1.17)
Median Age	0.12	0.11	0.12	0.12	0.12	0.12
	(0.11)	(0.11)	(0.11)	(0.11)	(0.11)	(0.11)
% Youth	–0.36*	–0.35*	–0.36*	–0.35*	–0.36*	–0.35*
	(0.15)	(0.15)	(0.15)	(0.15)	(0.15)	(0.15)
Fertility × % Youth	0.07	0.07	0.07	0.067	0.07	0.06
	(0.043)	(0.042)	(0.043)	(0.042)	(0.43)	(0.42)
Average Schooling Years, Females Over 15	0.27	—	—	—	—	—
	(0.27)					
Average Schooling Years, Females Over 25	—	0.48*	—	—	—	—
		(0.26)				
Average Schooling Years, Males Over 15	—	—	0.07	—	—	—
			(0.22)			
Average Schooling Years, Males Over 25	—	—	—	0.34	—	—
				(0.21)		
Average Schooling Years, Males + Females Over 15	—	—	—	—	0.18	—
					(0.26)	
Average Schooling Years, Males + Females Over 25	—	—	—	—	—	0.45*
						(0.23)
GDP	–0.000066	–0.000078	–0.000052	–0.00006	–0.00006	–0.000076
	(0.000046)	(0.000047)	(0.000046)	(0.00004)	(0.000046)	(0.000046)

Table 5 (continued)

Variable	Model 1 Females Over 15	Model 2 Females Over 25	Model 3 Males Over 15	Model 4 Males Over 25	Model 5 Males and Females Over 15	Model 6 Males and Females Over 25
Globalization	0.063*	0.05*	0.071*	0.057*	0.067*	0.052*
	(0.028)	(0.02)	(0.028)	(0.027)	(0.028)	(0.028)
History as French Colony	2.26	2.37	2.19	2.48	2.25	2.47
	(1.6)	(1.6)	(1.6)	(1.66)	(1.6)	(1.63)
Origin in British Common Law	3.45*	3.52*	3.48*	3.68*	3.48*	3.62*
	(1.66)	(1.65)	(1.66)	(1.66)	(1.66)	(1.66)
% Muslim	–0.036*	–0.033*	–0.039*	–0.038**	–0.038**	–0.036*
	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)
Constant	9.6	8.8	10.81	8.8	10.05	8.45
	(5.94)	(6.045)	(5.8)	(6.05)	(5.89)	(6.06)
N	621	614	621	614	621	614
Number of Groups	92	91	92	91	92	91
Observations per Group						
Minimum	2	2	2	2	2	2
Average	6.8	6.7	6.8	6.7	6.8	6.8
Maximum	7	7	7	7	7	7
Wald χ^2	10	10	10	10	10	10
df	388.04	397.61	373	370.34	377. 40	384.69
Prob > χ^2	.000	.000	.000	.000	.000	.000

Notes: All models display data controlling for average schooling years. Robust standard errors are shown in parentheses.

$^*p < .05$; $^{**}p < .01$; $^{***}p < .001$

(Przeworski et al. 2000), percentage of women using contraceptives, the CPI, and two separate measures for education (one in each of the two models in Table 4).

As the findings in Table 4 indicate, the inclusion of the additional controls decidedly limits the number of observations. Yet, the effect of fertility rate remains in the anticipated direction and is highly significant. Although the effect of the percentage of women using contraceptives is in the anticipated direction and marginally significant, the effect of fertility is still highly significant. The effect of CPI is in the anticipated direction and is highly significant, whereas the coefficient on GDP does not reach standard levels of statistical significance. Finally, the effects of the additional controls introduced in this model, such as the presence of neighboring democratic nations, is as expected based on the literature.

Table 5 presents a host of models testing for the effect of fertility with years of schooling controlled for. Here, too, support for the key hypothesis remains robust.

Discussion and Conclusions

The goal of this project is twofold: (1) to explain the effect of fertility rate on democracy even when controlling for the effects of traditional predictors of a democratic form of government, and (2) to demonstrate how this woman-related demographic variable directly influences political conditions above and beyond the effects of demographic variables already explored in the literature (Tickner 2005). The theoretical framework developed here and the empirical models estimated indicate an effect of fertility on democracy. This effect is reflected in a main effect for the fertility rate variable, notwithstanding model specification or the ways democracy is measured. This analysis makes significant theoretical and empirical contributions to our understanding of the connections between trends in the demos and in democracy. The connections between fertility rate and democracy are predicated on effects on women's political and economic empowerment, on changes in family structure, and possibly also on changes in legal conditions and the rule of law.

Controlling for alternative accounts for democracy—such as globalization, economic conditions, religious constituencies, and colonial heritage—and for the issue of causality between the key predictor and the outcome variable, I find that fertility rates systematically affect the extent to which a country is democratic. This finding holds not just for Indonesia, Nigeria, or Mexico, but it applies much more broadly. The inclusive list of countries in the sample suggests that the test for the key hypothesis is particularly stringent. Some countries in the sample sustained low fertility rates but still remain dictatorships (Madsen 2012). In countries where the reasons behind fertility decline differ from those in most other places where such a decline occurred in recent decades, we should not necessarily expect the same effect on democracy. This is true particularly in places such as China, where the regulation of fertility was part of social engineering and a state-building endeavor. Furthermore, deviant cases should not necessarily lead to refuting a thesis but rather be explained with reference to their own unique circumstances (Lipset 1959:70).

Beyond specific cases, given that most outlier countries where low fertility was not followed by democratization (Russia, China, Cuba, United Arab Emirates) are included in the sample (with North Korea as the only likely exception) and given that the effect of fertility is still highly significant in the range of models estimated, the test is stringent. The sample includes countries that belie the thesis;—for example, in the United Arab Emirates (UAE), fertility rates declined from 6.94 in 1950 to 1.77 in 2015, but democracy showed no noticeable change during that period. The fact that the key finding remains, although such countries are included in the sample, further underscores the overall strength of the thesis.

My findings suggest close proximity between demography and politics. When, on average, women have fewer children in their lifetime, the country in which they live is likely to become more democratic as a result, even if other conditions may push in the opposite direction (e.g., the potential depressing effect of Islam on democracy in Indonesia or the history of military regimes in this country). Furthermore, the causal direction from fertility rate to democracy is well established, notwithstanding whether fertility is lagged three years, five years, or entire political generations. In addition, the relationship between fertility and democracy is not a matter of economic modernity. Even in countries with low GDP per capita (see the appendix), the coefficient on fertility is negative and highly significant, as anticipated. Likewise, even when I control for economic conditions by specifying the CPI as one of the predictors, in addition to GDP per capita, the evidence for the link between fertility rate and democracy remains powerful. Furthermore, the results are robust to changes in model specification, alterations in data sources, and differences in measurement strategies for the different variables. Future research might explore the effects of fertility rate on other political variables, such as political violence (Urdal 2006).

My research sheds light on population planning programs at the national and international levels with ramifications that go beyond public health or medical policies. For instance, the findings here indicate that apart from their importance in fighting the spread of sexually transmitted diseases and their role in population planning, international programs promoting contraceptives and family planning may also bring about momentous political shifts. Inasmuch as this effect may be unintended, depressing fertility rates by increasing the use of contraceptives will likely also increase levels of democracy.[9]

As fertility rates worldwide are converging toward replacement level, this universal demographic trend is likely also to lead to democratization, even in corners of the globe where political and economic conditions were traditionally perceived as standing in the way of democratic reform. Recent political trends toward democracy in Indonesia, across the Arab world, and Latin America are prime examples. In those countries, fertility rates have fallen dramatically in recent decades, and recent political shifts should be analyzed with this in mind. Individuals' will to express their views, articulate their needs, freely form political groups, and in some cases act to overthrow dictators are at least partly accounted for by the fact that fertility rates have fallen in those countries from a level of approximately six children per woman to around replacement-level fertility rates over the course of the last three to four decades. This trend of the first demographic transition accounts for development in all parts of the globe. With fertility rates projected to decline considerably in Africa in the coming decades, sub-Saharan regions will be of particular interest for students of democracy and democratization.

[9] Even when I control for the effects of the percentage of women using contraceptives, which itself improves democratic conditions, the effect of decreasing fertility rates is still highly significant. Thus, the direct effect of contraceptive use on democracy is supplemented by the effect of decline in fertility rate on democratization.

Future research should further delve into some of the findings of this study. I find strong support for the hypothesis that female-related predictors have an independent effect on political structure. Yet, in the theoretical path model, fertility influences the four intervening variables—family structure, economic conditions of women, political empowerment of women, and land policy—which themselves link fertility causally to democracy. Within the scope of this article, I test an abridged version of the theoretical model. Still, future work should examine the effects of these intervening variables—not measured in the current analyses—to allow us a better understanding of the mechanisms at work linking demography and politic. Specifically, such future research would test the assumptions that the observed direct effect of fertility on democracy in fact results from the effect of fertility on family structure, land policy, women's economic conditions, and the political empowerment of women.

Via effects at the macro and micro (individual) levels on family structure, the empowerment of women, and land allocation, declines in fertility rates may lead in the direction of a democratic form of government, even when the obstacles for this type of development are high. Predictions concerning the spread of democracy based on population age structure (e.g., Cincotta 2009) and analysis of the relations between demography and democracy (e.g., Weber 2013) should be updated to reflect the independent and meaningful effect of women's demographics, most notably in the form of fertility rate.

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Appendix

Countries Included in the Samples

With minor variations, most multivariate models include most of the following countries: Albania, Algeria, Angola, Argentina, Armenia, Australia, Austria, Azerbaijan, Bahamas, Bahrain, Bangladesh, Barbados, Belarus, Belgium, Belize, Benin, Bolivia, Bosnia and Herzegovina, Botswana, Brazil, Bulgaria, Burkina Faso, Burundi, Cambodia, Cameroon, Canada, Cape Verde, Chad, Chile, China, Colombia, Costa Rica, Cote d'Ivoire, Croatia, Cuba, Cyprus, Denmark, Ecuador, Egypt, El Salvador, Estonia, Fiji, Finland, France, Gabon, Gambia, Georgia, Germany, Ghana, Greece, Grenada, Guatemala, Guinea-Bissau, Guyana, Haiti, Honduras, Hungary, Iceland, India, Indonesia, Iran, Ireland, Israel, Italy, Jamaica, Japan, Jordan, Kazakhstan, Kenya, Kuwait, Kyrgyzstan, Latvia, Lebanon, Lesotho, Lithuania, Luxembourg, Madagascar, Malawi, Malaysia, Maldives, Mali, Malta, Mauritania, Mauritius, Mexico, Moldova, Mongolia, Morocco, Mozambique, Myanmar, Namibia, Nepal, Netherlands, New Zealand, Nicaragua, Niger, Nigeria, Norway, Oman, Panama, Papua New Guinea, Paraguay, Peru, Philippines, Poland, Portugal, Qatar, Romania, Russian Federation, Rwanda, Samoa, Saudi Arabia, Senegal, Sierra Leone, Singapore, Slovenia, South Africa, Spain, Sri Lanka, Sudan, Suriname, Swaziland, Sweden, Switzerland, Syria, Tajikistan, Tanzania, Thailand, Togo, Trinidad and Tobago, Tunisia, Turkey, Uganda, Ukraine, United Arab Emirates, United Kingdom, United States, Uruguay, Vanuatu, Venezuela, Vietnam, Zambia, and Zimbabwe.

A Robustness Test: The Role of Economic Conditions and the Fertility-Democracy Link in Poor Nations

Economic conditions and democracy affect each other, with the effect of democracy on the economy changing over time (Krieckhaus 2004). Whereas some work has indicated that democracy influences economic performance (Almeida and Ferreira 2002), the literature provides some support for the opposite causal direction as well (Barro 1999). Likewise, fertility has been shown to be influenced by economic growth (Perotti 1996). Along with political conditions, a key predictor of fertility is the country's economy (Rouyer 1987). Hypothesis 1 states that the effect of fertility rate on democracy holds even when economic effects are controlled for. Furthermore, my theoretical frameworks flesh out how fertility rate may be independent from economic variables in influencing politics. To increase the robustness of this finding, I examine the fertility-democracy link for a sample that is limited to countries with minimal economic development. If this additional test yields findings that are in line with the hypotheses, this would add to the overall robustness of the conclusions.

To control for this alternative, I conduct analyses for poor countries only; results are presented in Table 6. Support for the key theoretical contention remains, demonstrating that fertility rate declines are associated with higher levels of democracy. The coefficient on fertility rate is negative, highly significant, and approximately of the same size as in Model 1 in Table 1. Even in a sample of countries with poor economic conditions, the effect of fertility on democracy is systematic and of a comparable magnitude to that in the universe of nations. Hypothesis 1, therefore, wins unequivocal support in this robustness test as well.

Table 6 Controlling for the effect of economic conditions (results for poor countries only)

Variable	Model
Fertility Rate	–1.35***
	(0.24)
GDP	0.00004
	(0.00007)
Globalization	0.045*
	(0.021)
History as French Colony	1.24
	(1.04)
Origin in British Common Law	3.43**
	(1.13)
% Muslim	–0.04***
	(0.01)
Other Democracies in Region	4.51**
	(1.55)
Constant	3.12*
	(1.8)

Table 6 (continued)

Variable	Model
N	3,496
Number of Groups	124
Observations Per Group	
Minimum	2
Average	28.2
Maximum	35
Wald χ^2	7
df	352.23
Prob > χ^2	0.000

Note: Robust standard errors are shown in parentheses.

*p < .05; **p < .01; ***p < .001

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